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Янин Василий Викторович

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МОДЕЛЕЙ**

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Научный руководитель
Доцент

А. Аланов

Студент

В.В. Янин

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Faculty of Computer Science

Vasily Yanin

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Supervisor

Aibek Alanov

Student

Vasily Yanin

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АННОТАЦИЯ

Работа посвящена задаче текстового редактирования изображений с использованием предобученных генеративных моделей. Основная проблема таких методов состоит в противоречии между силой семантического изменения и сохранением исходной структуры: глобальное смешивание векторных полей позволяет изменить объект, но одновременно искажает фон и неизменяемые области изображения. Цель исследования состоит в разработке и экспериментальной проверке подхода, который локализует редактирующее воздействие с помощью внутренних карт внимания модели. Для достижения цели рассмотрены современные методы Rectified Flow, Flow Matching, FlowEdit, оптимально-транспортные коррекции и методы выравнивания шума; сформулированы ограничения существующих подходов; предложен метод Spatially-Adaptive Velocity Fusion, в котором разность векторных полей источника и целевого описания умножается на пространственную маску, построенную по различию карт внимания. Эмпирическая часть выполнена на архитектуре Stable Diffusion 3 и подмножестве из 280 пар изображение-промпт. В качестве метрик использованы LPIPS, CLIP Score и HPSv2. Результаты показывают, что маскирование на основе внимания существенно улучшает сохранение структуры по LPIPS по сравнению с базовым алгоритмом FlowEdit, при этом ухудшение семантического соответствия остается небольшим. Лучший результат по LPIPS показывает soft-mask вариант, тогда как hard-mask вариант дает более сбалансированный компромисс между сохранением структуры, соответствием тексту и perceptual quality. Полученные результаты подтверждают, что внутренние карты внимания модели могут служить эффективным сигналом пространственного управления редактированием.

ABSTRACT

This thesis studies text-guided image editing with pre-trained flow-based generative models. The central difficulty of such methods is the trade-off between semantic edit strength and source-image preservation: global velocity blending can modify the target object, but it also affects the background and other regions that should remain unchanged. The goal of the work is to develop and experimentally evaluate an editing mechanism that localizes the editing force by using the model’s internal attention maps. To achieve this goal, the thesis reviews Rectified Flow, Flow Matching, FlowEdit, optimal-transport corrections, and noise-alignment methods; identifies unresolved limitations of existing approaches; and proposes Spatially-Adaptive Velocity Fusion. In the proposed method, the source-target velocity difference is multiplied by a spatial mask computed from attention-map differences, so that stronger editing is applied only to semantically relevant regions. The empirical evaluation is conducted on Stable Diffusion 3 using a subset of 280 image-prompt pairs. Editing quality is measured with LPIPS, CLIP Score, and HPSv2. The results show that attention-based masking substantially improves structural preservation according to LPIPS compared with the FlowEdit baseline, while the loss in semantic alignment remains small. The soft-mask variant achieves the best LPIPS score, whereas the hard-mask variant offers a more balanced trade-off between structure preservation, text alignment, and perceptual preference. Overall, the experiments support the hypothesis that internal attention maps of a flow-based text-to-image model can provide an effective spatial control signal for localized image editing.

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INTRODUCTION

Text-guided image editing has become one of the most practically important applications of modern generative models. A user provides an input image and a natural-language instruction, while the model is expected to change only the semantically relevant part of the image and preserve all remaining content. This formulation is useful for creative design, visual prototyping, data augmentation, and interactive media production, but it is technically difficult because semantic changes and structural preservation often require conflicting transformations of the latent representation.

Recent progress in text-to-image generation has shifted part of the field from stochastic diffusion models to deterministic flow-based models. Rectified Flow and Flow Matching provide a convenient continuous-time view of generation as transport between a noise distribution and a data distribution. In image editing, these models make it possible to operate directly with velocity fields conditioned on the source and target prompts. However, the commonly used global blending of source and target velocities is not spatially selective: the same editing force is applied to the whole latent tensor, even when only one object should be changed.

The relevance of this thesis is determined by this unresolved localization problem. Existing flow-based editing methods can preserve global image structure, but they often under-edit the target object when the guidance is weak and corrupt background details when the guidance is strong. Therefore, improving spatial control in velocity-based editing is essential for making such methods more reliable in realistic editing scenarios.

The object of this research is text-guided image editing with pre-trained generative models. The subject of the research is the construction and modulation of velocity fields in flow-based editing algorithms.

The goal of the thesis is to develop and evaluate an attention-based modification of FlowEdit that improves structure preservation by applying the editing velocity primarily to semantically relevant regions.

To achieve this goal, the following tasks are addressed:

1. analyze the theoretical foundations of Rectified Flow, Flow Matching, and modern text-guided image editing methods;
2. identify the limitations of global velocity blending in inversion-free flow-based

editing;

3. formulate several hypotheses for improving the editability–preservation trade-off;
4. develop the Spatially-Adaptive Velocity Fusion method based on attention-derived masks;
5. implement hard-mask and soft-mask variants of the method on top of Stable Diffusion 3 [3];
6. evaluate the proposed variants against the FlowEdit baseline and a velocity-gradient correction baseline using LPIPS [8], CLIP Score [7], and HPSv2 [9];
7. summarize the empirical findings and define directions for further research.

The methodological basis of the work includes continuous-time generative modeling, neural ordinary differential equations, attention mechanisms in transformer-based diffusion models, and empirical evaluation of image editing quality. The empirical base consists of 280 source-image and editing-prompt pairs from the public evaluation dataset that the FlowEdit authors collected and released alongside their official implementation [4].

The main result of the thesis is an experimentally validated attention-masked velocity fusion procedure. The proposed method reduces LPIPS from 0.2129 for the baseline to 0.1415 in the best soft-mask configuration, indicating substantially better structural preservation while retaining competitive semantic alignment.

The thesis is structured as follows. Chapter 1 reviews the related work on inversion-free and inversion-based flow editing and identifies the research gap. Chapter 2 introduces the preliminaries: the rectified-flow and flow-matching formulations on which the work builds, and a formal statement of the editing task addressed by the thesis. Chapter 3 describes the proposed method, including Spatially-Adaptive Velocity Fusion, the attention-difference mask construction, and Velocity-Gradient Correction. Chapter 4 presents the experimental setup, the quantitative sweeps, and the qualitative case studies. The conclusion summarizes the contributions and discusses future research directions.

1 RELATED WORK

1.1 Inversion-Free Flow-Based Editing

Classical editing methods often invert a real image into a latent noise representation and then run a modified generation process. This inversion step is fragile because numerical errors accumulate over many discretized timesteps. Inversion-free approaches avoid explicitly mapping the image back to pure noise and instead construct an editing trajectory directly.

FlowEdit [4] is a representative inversion-free method for text-guided editing with pre-trained flow models. It defines an editing latent Z_t^{FE} , initialized at the source latent X^{src} , and integrates it directly under the velocity *difference* between the target- and source-conditioned fields:

$$Z_{t_{i-1}}^{\text{FE}} = Z_{t_i}^{\text{FE}} + (t_{i-1} - t_i)(V_t^{\text{tar}} - V_t^{\text{src}}). \quad (1.1)$$

The two velocities are evaluated at noise-displaced positions $Z_t^{\text{src}} = (1 - t)X^{\text{src}} + t\epsilon$ and $Z_t^{\text{tar}} = Z_t^{\text{FE}} + Z_t^{\text{src}} - X^{\text{src}}$, where a fresh $\epsilon \sim \mathcal{N}(0, I)$ is sampled at every step. Crucially, the editing latent is never integrated under V^{src} on its own — only under the prompt-induced delta $V_t^\Delta = V_t^{\text{tar}} - V_t^{\text{src}}$. The method is attractive because it does not require training a task-specific model and can be applied to modern text-to-image architectures.

However, FlowEdit relies on a global velocity correction. The same velocity difference is applied across the entire latent representation, even if the requested edit concerns only one object or a small semantic region. This leads to three important limitations. First, strong edits can corrupt background details because the background receives the same global correction as the target object. Second, weak edits preserve the background but may fail to change the object sufficiently. Third, global blending is not guaranteed to follow an optimal or stable transport path when the source and target prompts imply different geometry.

Optimal-transport corrections for rectified-flow editing [5] address part of the trajectory optimality problem by adding a transport-aware direction to the velocity field. Such methods improve the geometric regularity of the path, but they do not directly solve spatial localization: minimizing a global transport cost may still penalize

large object-level changes that are semantically necessary for editing.

1.2 Inversion-Based and Noise-Alignment Approaches

A second family of methods explicitly inverts the real image into a latent noise representation and then re-runs a modified generation process. Their main weakness is accumulated drift: small numerical errors in the reverse trajectory move the latent away from the source manifold. DNAEdit [6] mitigates this through Direct Noise Alignment, which refines randomly initialized noise in the Gaussian domain using velocity-field corrections, augmented by a Mobile Velocity Guidance schedule. Such techniques reduce reconstruction error, but at the cost of an additional optimization in noise space and a guidance schedule that is sensitive to hyperparameters. These trade-offs motivate train-free, inversion-free methods that preserve the simplicity of velocity-based editing while improving spatial selectivity, which is the direction taken by this work.

1.3 Research Gap

The reviewed literature converges on two unresolved challenges that directly motivate the present work. First, the editability / preservation trade-off: strong semantic edits damage the background, while conservative edits fail to change the target object. Second, the lack of spatial localization: existing flow-based editing methods operate on the velocity field as a whole and do not explicitly identify the regions that should and should not be edited. Transport-based and inversion-based approaches address related issues (path regularity, reconstruction error) but introduce additional complexity without resolving the spatial-selectivity problem itself. This thesis therefore focuses on adding an explicit spatial control signal to the FlowEdit update.

1.4 Chapter Conclusions

The chapter situated the work within inversion-free flow-based editing and identified the lack of spatial selectivity in global velocity blending as the open problem that motivates the proposed method.

2 PRELIMINARIES

2.1 Flow-Based Generative Modeling

The theoretical foundation for modern train-free image editing lies in continuous-time generative modeling. Rectified Flow [2] introduces a deterministic ordinary-differential-equation framework for transferring samples between two distributions:

$$dZ_t = v_\theta(Z_t, t) dt, \quad Z_0 \sim p_0, \quad Z_1 \sim p_1, \quad (2.1)$$

where p_0 is an initial distribution, p_1 is a target distribution, and v_θ is a neural vector field. The method learns trajectories that are close to straight lines between samples from the two distributions. A typical training objective has the form

$$\int_0^1 \mathbb{E} \left[\|(X_1 - X_0) - v_\theta(X_t, t)\|^2 \right] dt, \quad (2.2)$$

where $X_t = tX_1 + (1 - t)X_0$. The main motivation is computational efficiency: straighter paths can be simulated with fewer discretization steps.

Flow Matching [1] extends this idea by introducing a simulation-free training objective for continuous normalizing flows. Instead of solving an ODE during training, it regresses a neural vector field against conditional probability paths:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, q(x_1), p_t(x|x_1)} \|v_t(x) - u_t(x|x_1)\|^2. \quad (2.3)$$

This formulation enables scalable training and provides a direct connection between generative modeling and transport between distributions.

For image editing, the relevance of these frameworks is that generation can be interpreted as movement through a latent space under a learned velocity field. If the velocity field is conditioned on text, then changing the conditioning text changes the direction of the trajectory. This makes it possible to formulate editing as a controlled modification of the velocity field rather than as a separate optimization problem.

2.2 Problem Formulation

The thesis addresses the following task. Given a source image X^{src} , a source text prompt p^{src} that describes the content of X^{src} , and a target text prompt p^{tar} that describes the content of the desired edited image, the goal is to produce an edited image X^{tar} that

- (i) is *semantically aligned* with the target prompt p^{tar} wherever the two prompts disagree, and
- (ii) *preserves* the content of the source image X^{src} at all other spatial positions.

We restrict attention to the train-free, inversion-free setting: no task-specific training is performed, no inversion of X^{src} into a Gaussian noise representation is required, and the only available signal is the velocity field of a pre-trained text-to-image flow model conditioned on the two prompts.

Within the continuous-time formulation of Section 2.1, the source- and target-conditioned velocity fields at time t are denoted $V_t^{\text{src}} = v_\theta(z, t, c^{\text{src}})$ and $V_t^{\text{tar}} = v_\theta(z, t, c^{\text{tar}})$, where z is a latent and $c^{\text{src}}, c^{\text{tar}}$ are the text embeddings of p^{src} and p^{tar} . The prompt-induced velocity difference at the same point is

$$V_t^\Delta = V_t^{\text{tar}} - V_t^{\text{src}}, \quad (2.4)$$

and existing inversion-free flow-editing methods (Section 1.1) integrate the editing latent along the trajectory induced by V_t^Δ . The central design question of this thesis is then: *how should the prompt-induced velocity difference V_t^Δ be modulated so that the resulting trajectory satisfies (i) and (ii) simultaneously?* The proposed method (Chapter 3) answers this question with two complementary spatial-modulation schemes, evaluated quantitatively against the editability/preservation trade-off described in the experimental setup (Section 4.1).

2.3 Chapter Conclusions

The chapter introduced the continuous-time flow framework on which the proposed method builds and formalized the editing task as a controlled modulation of the prompt-induced velocity difference under joint editability and preservation constraints.

3 PROPOSED METHOD

3.1 Overview

The proposed method adds an explicit spatial control signal to the FlowEdit update introduced in Section 1.1. The core research question of the thesis — can internal attention maps of a text-to-image flow model serve as a spatial control signal for velocity-based editing? — is grounded in the observation that transformer attention maps encode which latent positions and which tokens are involved in representing objects and attributes. If an editing instruction changes only a localized object, the attention difference between source and target conditions should therefore provide a useful proxy for the region where the velocity field must be modified.

Two complementary modifications of FlowEdit are proposed and evaluated. Both keep the editing-latent integration scheme intact and modify only how the right-hand side of the update is constructed. The first, Spatially-Adaptive Velocity Fusion (Section 3.2), is the main method: it gates the source–target velocity difference by an attention-derived spatial mask whose construction is detailed in Section 3.3. The second, Velocity-Gradient Correction (Section 3.4), is a peer method that does not use attention maps and instead augments the velocity difference using its own gradient with respect to the target-conditioned latent; it serves both as an independent variant and as a control that isolates the contribution of spatial selectivity in the main method. Pseudocode for both attention-masked variants is given in Section 3.5.

3.2 Spatially-Adaptive Velocity Fusion

Spatially-Adaptive Velocity Fusion replaces the uniform velocity difference of FlowEdit with a spatially varying gate. The editing-latent update of FlowEdit is preserved, but the prompt-induced delta is masked element-wise before integration:

$$Z_{t_{i-1}}^{\text{FE}} = Z_{t_i}^{\text{FE}} + (t_{i-1} - t_i) M_{t_i} \odot (V_t^{\text{tar}} - V_t^{\text{src}}), \quad (3.1)$$

where $M_{t_i}(x) \in [0, 1]$ is a dynamic mask and $\odot$ denotes element-wise multiplication. The mask is extracted from the difference between source-conditioned and target-conditioned attention maps.

The intended effect is to decouple the foreground edit from background preser-

vation. With $M \equiv 1$ the update reduces to the FlowEdit baseline (the editing latent receives the full delta at every position); with $M \equiv 0$ the update vanishes and Z^{FE} stays at X^{src} . A high mask value therefore allows a strong editing force in semantically relevant regions, while a low mask value suppresses the edit in regions that should remain unchanged.

3.3 Attention-Difference Mask Construction

Both mask variants are derived from the same per-position scalar field $\hat{\Delta}(x) \in [0, 1]$, the *attention-difference map*, which records how differently each latent position attends to the source versus to the target text condition. The construction proceeds in three stages: extract per-block attention maps, aggregate them into a 2D field, then turn the field into either a hard or a soft mask.

Per-block attention maps. The MM-DiT backbone [3] computes, in every block ℓ , a joint self-attention over image and text tokens. We intercept the image-query / text-key sub-block of these probabilities and average it over attention heads and text tokens; the result is a scalar map over image positions, $m^{(\ell)}(x)$. This is done for the source-conditioned and target-conditioned forward passes of the current sampling step, yielding $m_{\text{src}}^{(\ell)}(x)$ and $m_{\text{tar}}^{(\ell)}(x)$. Every block of the transformer is used; no layer-subset selection is performed.

Per-step attention difference. The two per-block scalar maps are subtracted point-wise, taken in absolute value, and averaged across blocks:

$$\bar{\Delta}(x) = \frac{1}{N_{\text{blk}}} \sum_{\ell=1}^{N_{\text{blk}}} |m_{\text{tar}}^{(\ell)}(x) - m_{\text{src}}^{(\ell)}(x)|. \quad (3.2)$$

The attention buffer is reset between sampling steps, so $\bar{\Delta}$ is built only from the current step.

Spatial post-processing. The 1D vector $\bar{\Delta}$ is then (i) clipped from above at its 99th empirical percentile to suppress isolated outliers; (ii) reshaped to the 2D latent grid and bilinear-upsampled to its spatial resolution if needed; (iii) min-max normalized per step into $[0, 1]$. The result is the attention-difference map $\hat{\Delta}(x)$.

Hard and soft masks. The hard mask is the indicator that $\hat{\Delta}$ (after a small spatial smoothing by a 5×5 box average, Pool) exceeds a threshold; the soft mask is a shifted-and-scaled sigmoid of $\hat{\Delta}$, smoothed by the same pool:

$$M^{\text{hard}}(x) = \mathbf{1}[(\text{Pool } \hat{\Delta})(x) > \tau], \quad M^{\text{soft}}(x) = \text{Pool}(s \cdot \sigma((\hat{\Delta} - b)g))(x). \quad (3.3)$$

Defaults are $\tau = b = 0.15$, $g = 10$, $s = 1$; the threshold/bias sweeps used in the experiments are $\tau, b \in \{0.09, 0.12, 0.15, 0.18, 0.21\}$ (Tables 4.1, 4.2). The order of operations differs by design: for the hard mask, smoothing is applied to $\hat{\Delta}$ directly so that the threshold sees a denoised score; for the soft mask the sigmoid is what produces the continuous values, so smoothing is applied to its output to suppress high-frequency speckle.

3.4 Velocity-Gradient Correction

Velocity-Gradient Correction also keeps the FlowEdit integration scheme intact and modifies only the right-hand side of the editing-latent update, but the modification is driven by the velocity field itself rather than by attention maps. The motivation is that the prompt-induced delta $V_t^\Delta = V_t^{\text{tar}} - V_t^{\text{src}}$ is the result of a cancellation between two classifier-free-guided velocity predictions, and may carry only a fraction of the signal that actually distinguishes the target prompt from the source prompt. A small additional correction in the direction in which this difference grows fastest could amplify the editing signal without changing the integration rule. Concretely, the method computes the gradient of $\|V_t^\Delta\|_2^2$ with respect to the target-conditioned latent Z_t^{tar} :

$$g_t = \nabla_{Z_t^{\text{tar}}} \|V_t^{\text{tar}} - V_t^{\text{src}}\|_2^2. \quad (3.4)$$

The raw gradient g_t can have an arbitrary magnitude that depends on local curvature of the velocity field and on the scale of the model output, so it is first *rescaled* to match the norm of the delta velocity itself and then *clipped* element-wise:

$$\tilde{g}_t = \text{clip}_{[-\tau, \tau]} \left(\frac{\|V_t^\Delta\|}{\|g_t\| + \varepsilon} g_t \right). \quad (3.5)$$

The correction is added to V_t^Δ with a time-decaying coefficient $\gamma_t = \beta \cdot t/t_{\max}$ to produce the enhanced velocity:

$$V_t^{\text{enh}} = V_t^\Delta + \gamma_t \tilde{g}_t. \quad (3.6)$$

The editing-latent update then uses the FlowEdit step with V_t^{enh} in place of V_t^Δ :

$$Z_{t_{i-1}}^{\text{FE}} = Z_{t_i}^{\text{FE}} + (t_{i-1} - t_i) V_t^{\text{enh}}. \quad (3.7)$$

With $\beta = 0$ the correction vanishes ($\gamma_t \equiv 0$) and $V_t^{\text{enh}} = V_t^\Delta$, so the update reduces exactly to the FlowEdit baseline. For small positive β , the editing trajectory is pushed in the direction in which the velocity difference grows fastest, which empirically increases preservation in regions where the prompt-induced signal is already strongest. The time-decay factor $t/t_{\max}$ concentrates the correction on early sampling steps (when the latent is far from data and a global push is safe) and gradually removes it as the latent approaches the data manifold (when an unselective push would damage fine details). Unlike the attention-mask construction of Section 3.3, Velocity-Gradient Correction does not require access to internal attention maps and uses only the velocity outputs of the model; on the other hand, the resulting signal is a per-element scalar field derived from a single global objective and therefore cannot impose an explicit “edit-here, preserve-there” decomposition.

3.5 Algorithms

Algorithm 1 and Algorithm 2 summarize the hard-mask and soft-mask variants of Spatially-Adaptive Velocity Fusion. Both follow the same editing-latent integration scheme as the FlowEdit baseline of Section 1.1; they differ only in how the attention-derived gate is constructed from the per-step attention-difference map $\widehat{\Delta A}_{t_i}$ of Section 3.3. For brevity the pseudocode collapses the construction of $\widehat{\Delta A}$ into a single `Normalize(Clip99(.))` step; the full construction (per-block reduction, source–target difference, clipping, reshape, normalization) is the one described in Section 3.3. Velocity-Gradient Correction follows the same scheme with the same outer loop but uses the enhanced velocity V_t^{enh} of Section 3.4 in place of the masked delta; we do not duplicate its pseudocode.

Algorithm 1 Attention-Masked FlowEdit with Hard Mask

Require: Source image X^{src} , source prompt p_{src} , target prompt p_{tar} , timesteps $\{t_i\}_{i=0}^T$, threshold τ

Ensure: Edited image X^{tar}

- 1: **Initialize** $Z_{t_{\max}}^{\text{FE}} \leftarrow X^{\text{src}}$
 - 2: **for** $i = t_{\max}$ **down to** 1 **do**
 - 3: Sample $N_{t_i} \sim \mathcal{N}(0, I)$
 - 4: Construct $Z_{t_i}^{\text{src}} \leftarrow (1 - t_i)X^{\text{src}} + t_i N_{t_i}$
 - 5: Construct $Z_{t_i}^{\text{tar}} \leftarrow Z_{t_i}^{\text{FE}} + Z_{t_i}^{\text{src}} - X^{\text{src}}$
 - 6: Extract A^{src} from $V^{\text{src}}(Z_{t_i}^{\text{src}}, t_i, p_{\text{src}})$
 - 7: Extract A^{tar} from $V^{\text{tar}}(Z_{t_i}^{\text{tar}}, t_i, p_{\text{tar}})$
 - 8: $\Delta A_{t_i} \leftarrow |A^{\text{src}} - A^{\text{tar}}|$
 - 9: $\widehat{\Delta A}_{t_i} \leftarrow \text{Normalize}(\text{Clip99}(\Delta A_{t_i}))$
 - 10: $M_{t_i} \leftarrow \mathbf{1}(\widehat{\Delta A}_{t_i} > \tau)$
 - 11: $V_{t_i}^{\Delta} \leftarrow M_{t_i} \odot (V^{\text{tar}} - V^{\text{src}})$
 - 12: $Z_{t_{i-1}}^{\text{FE}} \leftarrow Z_{t_i}^{\text{FE}} + (t_{i-1} - t_i)V_{t_i}^{\Delta}$
 - 13: **end for**
 - 14: **return** $Z_0^{\text{FE}} = X^{\text{tar}}$
-

Algorithm 2 Attention-Masked FlowEdit with Soft Mask

Require: Source image X^{src} , source prompt p_{src} , target prompt p_{tar} , timesteps $\{t_i\}_{i=0}^T$, parameters $\{bias, gain, scale\}$

Ensure: Edited image X^{tar}

- 1: **Initialize** $Z_{t_{\text{max}}}^{\text{FE}} \leftarrow X^{\text{src}}$
 - 2: **for** $i = t_{\text{max}}$ **down to** 1 **do**
 - 3: Sample $N_{t_i} \sim \mathcal{N}(0, I)$
 - 4: Construct $Z_{t_i}^{\text{src}} \leftarrow (1 - t_i)X^{\text{src}} + t_i N_{t_i}$
 - 5: Construct $Z_{t_i}^{\text{tar}} \leftarrow Z_{t_i}^{\text{FE}} + Z_{t_i}^{\text{src}} - X^{\text{src}}$
 - 6: Extract A^{src} from $V^{\text{src}}(Z_{t_i}^{\text{src}}, t_i, p_{\text{src}})$
 - 7: Extract A^{tar} from $V^{\text{tar}}(Z_{t_i}^{\text{tar}}, t_i, p_{\text{tar}})$
 - 8: $\Delta A_{t_i} \leftarrow |A^{\text{src}} - A^{\text{tar}}|$
 - 9: $\widehat{\Delta A}_{t_i} \leftarrow \text{Normalize}(\text{Clip99}(\Delta A_{t_i}))$
 - 10: $\alpha_{t_i} \leftarrow scale \cdot \sigma((\widehat{\Delta A}_{t_i} - bias) \cdot gain)$
 - 11: $V_{t_i}^{\Delta} \leftarrow \alpha_{t_i} \odot (V^{\text{tar}} - V^{\text{src}})$
 - 12: $Z_{t_{i-1}}^{\text{FE}} \leftarrow Z_{t_i}^{\text{FE}} + (t_{i-1} - t_i)V_{t_i}^{\Delta}$
 - 13: **end for**
 - 14: **return** $Z_0^{\text{FE}} = X^{\text{tar}}$
-

3.6 Chapter Conclusions

The chapter introduced Spatially-Adaptive Velocity Fusion, which gates the prompt-induced velocity difference by an attention-derived spatial mask, and Velocity-Gradient Correction, which augments the same velocity difference using its own gradient with respect to the target-conditioned latent. Both methods preserve the FlowEdit integration scheme exactly and differ only in how the right-hand side of the update is built. The next chapter evaluates them empirically.

4 EXPERIMENTAL VALIDATION

4.1 Experimental Setup

All methods were implemented on top of the Stable Diffusion 3 architecture by modifying the FlowEdit sampling procedure. The evaluation was conducted on the 280 source-image and editing-prompt pairs of the public benchmark released by the FlowEdit authors, which they assembled specifically for that work and published on the project GitHub repository together with the official implementation of the method [4]. The benchmark includes object replacement, attribute modification, and localized semantic edits.

The experiments used a fixed random seed and the following standardized hyperparameters:

- diffusion steps: $T = 50$;
- averaging samples: $n_{\text{avg}} = 1$;
- source classifier-free guidance scale: $\gamma_{\text{src}} = 3.5$;
- target classifier-free guidance scale: $\gamma_{\text{tar}} = 13.5$.

Editing quality was evaluated using three complementary metrics:

1. **LPIPS** [8] measures perceptual distance between the source and edited images. Lower values indicate better preservation of source structure and background.
2. **CLIP Score** [7] measures semantic alignment between the edited image and the target prompt. Higher values indicate stronger text-image consistency.
3. **HPSv2** [9] is a learned human-preference score that estimates overall visual and semantic quality. Higher values indicate better predicted perceptual preference.

Four methods are compared throughout the chapter: the FlowEdit baseline of Section 1.1, the two attention-masked variants of Spatially-Adaptive Velocity Fusion (Section 3.2) — with hard and soft mask (Algorithms 1–2) — and Velocity-Gradient Correction (Section 3.4). All four share the editing-latent integration scheme and differ only in how the right-hand side of the update is constructed.

4.2 Quantitative Results

4.2.1 Attention-Masked Variants

This section reports a controlled sweep of the mask-strength parameter for both masked variants. For the hard mask the parameter is the binarization threshold τ applied to the normalized attention difference map; for the soft mask the parameter is the sigmoid bias b that shifts the activation point (with gain = 10 and scale = 1 fixed). All other settings are identical to the baseline ($T = 50$, $n_{\min} = 0$, $n_{\max} = 33$, CFG 3.5/13.5). All values are means over the same 280 image-prompt pairs.

Hard-mask threshold sweep

Table 4.1 reports the hard-mask variant over the threshold grid $\tau \in \{0.09, 0.12, 0.15, 0.18, 0.21\}$ together with the FlowEdit baseline.

Table 4.1. Hard-mask threshold sweep against the FlowEdit baseline (280 images).

Method	τ	LPIPS ↓	CLIP Score ↑	HPSv2 ↑
FlowEdit baseline	—	0.2129	35.0897	0.2970
Hard mask	0.09	0.1922	35.1057	0.2960
Hard mask	0.12	0.1763	35.0759	0.2954
Hard mask	0.15	0.1617	35.0205	0.2946
Hard mask	0.18	0.1502	34.9331	0.2937
Hard mask	0.21	0.1402	34.8672	0.2924

As τ grows from 0.09 to 0.21, the active mask region shrinks to the strongest attention-difference pixels. This monotonically improves preservation (LPIPS drops from 0.1922 to 0.1402, a 34–45% reduction relative to the 0.2129 baseline) at the cost of a small but monotone decrease of text alignment (CLIP from 35.1057 to 34.8672, -0.24 points, $\approx 0.7\%$) and a marginal drop of perceptual quality (HPSv2 from 0.2960 to 0.2924, -0.0036). The smallest threshold $\tau = 0.09$ actually *exceeds* the baseline on CLIP (35.1057 vs. 35.0897) while already improving LPIPS by 9.7%, which shows that even a very permissive mask is non-trivially better than no mask at all. The point $\tau = 0.15$ sits at the elbow of the curve: it removes roughly half of the baseline LPIPS gap while keeping CLIP and HPSv2 within 0.07 and 0.0024 of the

baseline, respectively.

Soft-mask bias sweep

Table 4.2 reports the soft-mask variant over the sigmoid-bias grid $b \in \{0.09, 0.12, 0.15, 0.18, 0.21\}$ together with the FlowEdit baseline.

Table 4.2. Soft-mask bias sweep against the FlowEdit baseline (280 images).

Method	b	LPIPS ↓	CLIP Score ↑	HPSv2 ↑
FlowEdit baseline	—	0.2129	35.0897	0.2970
Soft mask	0.09	0.1584	34.9189	0.2941
Soft mask	0.12	0.1501	34.8757	0.2934
Soft mask	0.15	0.1415	34.8423	0.2927
Soft mask	0.18	0.1330	34.8096	0.2918
Soft mask	0.21	0.1246	34.7691	0.2908

The soft mask follows the same qualitative pattern as the hard one but is more aggressive at every operating point. Increasing b from 0.09 to 0.21 lowers LPIPS from 0.1584 to 0.1246 (a further 41% reduction over the baseline) but the CLIP penalty stays small ($34.9189 \rightarrow 34.7691$, -0.15 points, $\approx 0.4\%$) and HPSv2 loses only 0.0033. Comparing the two variants at matched parameter values, the soft mask consistently produces a lower LPIPS than the hard mask (e.g. at 0.21: 0.1246 vs. 0.1402; at 0.15: 0.1415 vs. 0.1617) while paying a slightly larger CLIP cost. This is consistent with the geometric intuition: the soft mask attenuates the velocity difference smoothly across the entire latent grid, suppressing low-amplitude background updates that the hard mask would still pass through unchanged once they exceed τ .

Best per type vs. baseline

For each variant we select the operating point most favourable to the variant’s typical use case. For the hard mask we choose $\tau = 0.09$, which is the unique configuration that improves LPIPS *and* improves CLIP Score relative to the baseline — i.e. Pareto-dominates the baseline on two of the three metrics while losing only 0.001 on HPSv2. For the soft mask such a Pareto-dominant point does not exist (CLIP is below the baseline at every value of b in the sweep), so we fall back to the LPIPS-minimum

point $b = 0.21$, which delivers the strongest structure preservation. The resulting two operating points are compared with the baseline in Table 4.3.

Table 4.3. Best operating point of each mask type against the FlowEdit baseline.

Method	Param	LPIPS ↓	CLIP Score ↑	HPSv2 ↑
FlowEdit baseline	—	0.2129	35.0897	0.2970
Hard mask (Pareto-dominant)	$\tau = 0.09$	0.1922	35.1057	0.2960
Soft mask (best LPIPS)	$b = 0.21$	0.1246	34.7691	0.2908

Trade-offs of the best-of-each comparison.

- **Hard mask is a free lunch on preservation.** Relative to the baseline, the hard mask at $\tau = 0.09$ reduces LPIPS by 9.7% ($0.2129 \rightarrow 0.1922$) while simultaneously *improving* CLIP Score by 0.016 points ($35.0897 \rightarrow 35.1057$), and losing only 0.0010 on HPSv2 (-0.34%). For all practical purposes this is a strict improvement over the baseline: there is no metric on which the change is meaningfully negative.
- **Soft mask trades alignment for preservation aggressively.** The soft mask at $b = 0.21$ reduces LPIPS by 41.5% ($0.2129 \rightarrow 0.1246$) at the cost of -0.32 CLIP points (-0.9%) and -0.0062 HPSv2 (-2.1%). This is a much larger preservation gain than the hard mask, but it has to be paid for in semantic alignment and human preference. Both losses are still sub-percent to low single-digit, so the trade remains favourable for preservation-dominated use cases.
- **Hard vs. soft.** The two variants occupy fundamentally different points on the Pareto front: the hard mask at $\tau = 0.09$ is a strict improvement over the baseline (positive on LPIPS *and* CLIP), whereas the soft mask at $b = 0.21$ is a Pareto trade (large LPIPS gain, smaller CLIP/HPSv2 loss). Use the hard mask when the editing pipeline cannot tolerate any regression on semantic alignment or human-rated quality (e.g. prompt-faithfulness benchmarks). Use the soft mask when pixel- and patch-level fidelity to the source matters more than the last decimal of CLIP / HPSv2 (e.g. inpainting-style edits, identity preservation).
- **Parameter sensitivity.** Both metrics in both sweeps are smooth and monotone in the mask-strength parameter, with no sign of non-monotonicity or in-

stability across the studied range. The threshold/bias is therefore a clean and predictable knob: a practitioner can dial in the desired preservation/alignment balance simply by moving along the sweep curve, without rerunning hyperparameter search. Concretely, for the hard mask increasing τ from 0.09 to 0.21 buys an extra 27% LPIPS reduction in exchange for the small CLIP/HPSv2 advantage observed at $\tau = 0.09$.

- **Sweet spot.** If a single “default” is required, the value 0.15 is a reasonable elbow choice for both variants: it captures about 70–75% of the LPIPS improvement available at 0.21 while keeping CLIP within 0.07–0.25 of the baseline. The value 0.09 is the recommended choice when no regression on CLIP is acceptable. The value 0.21 is the recommended choice when the goal is maximal LPIPS preservation.

4.2.2 Velocity-Gradient Correction

Velocity-Gradient Correction (VGC) augments the prompt-induced velocity difference with a rescaled-and-clipped gradient correction weighted by a time-decaying coefficient; the full formulation is given in Section 3.4. The hyper-parameter β controls how strongly the gradient correction is applied; in this sweep we keep the clip threshold at $\tau = 1.0$ and vary $\beta \in \{0.1, 0.3, 0.5\}$. Larger values of β (tested up to 5.0) push the latent outside the useful operating range and degrade every metric, so we restrict the reported sweep to the working interval. All other settings are identical to the baseline. Table 4.4 reports the full sweep.

Table 4.4. Velocity-Gradient Correction sweep over β against the FlowEdit baseline (280 images, $\tau = 1.0$).

Method	β	LPIPS ↓	CLIP Score ↑	HPSv2 ↑
FlowEdit baseline	—	0.2129	35.0897	0.2970
VGC	0.1	0.2099	35.0583	0.2962
VGC	0.3	0.2054	35.0570	0.2946
VGC	0.5	0.2022	35.0722	0.2928

- **Modest monotone preservation gain.** In the working interval $\beta \in [0.1, 0.5]$, LPIPS decreases monotonically from 0.2099 to 0.2022, corresponding to a

1.4%–5.0% relative improvement over the baseline. The effect is real but small compared with the masked variants.

- **CLIP Score is essentially flat.** Across the operating range CLIP stays within 0.03 of the baseline (35.0570–35.0722 vs. 35.0897). VGC therefore does not visibly trade semantic alignment for preservation in this regime — the gradient amplification is too global to selectively change content versus background.
- **HPSv2 is the most sensitive metric.** HPSv2 decreases monotonically with β , from 0.2962 at $\beta = 0.1$ (−0.0008) to 0.2928 at $\beta = 0.5$ (−0.0042). Even at the best LPIPS point, HPSv2 is already 0.14%–1.4% lower than the baseline. Outside the reported range ($\beta \geq 1$), HPSv2 collapses by an order of magnitude faster than LPIPS or CLIP, which makes it a reliable early-stopping signal for tuning β .
- **Best operating point vs. masked variants.** The minimum-LPIPS configuration is $\beta = 0.5$: LPIPS 0.2022, CLIP 35.0722, HPSv2 0.2928. This is the operating point used for VGC in the consolidated comparison (Section 4.5). Compared with the best attention-masked variants from Table 4.3, VGC delivers only $\approx 1/8$ of the LPIPS improvement that the soft mask delivers (0.0107 vs. 0.0883), at a similar HPSv2 cost. This quantitatively confirms the qualitative claim that velocity-gradient amplification cannot replace spatial selectivity.

4.2.3 Summary Comparison

Table 4.5 provides a consolidated comparison of the evaluated methods.

Table 4.5. Summary comparison of evaluated methods on 280 images.

Method	Param	LPIPS ↓	CLIP Score ↑	HPSv2 ↑
FlowEdit baseline	—	0.2129	35.09	0.2970
Soft mask	$b = 0.15$	0.1415	34.84	0.2927
Hard mask	$\tau = 0.09$	0.1922	35.11	0.2960
Velocity-Gradient Correction	$\beta = 0.5$	0.2022	35.07	0.2928

The summary comparison shows that attention masking is the most effective modification among the tested strategies. The soft mask is preferable when structural

preservation is the primary objective. The hard mask at $\tau = 0.09$ Pareto-dominates the baseline on (LPIPS, CLIP) with only a 0.001 loss on HPSv2 and is therefore the safest drop-in replacement when no regression on semantic alignment is acceptable. Velocity-Gradient Correction provides only a marginal preservation gain and does not offer a compelling overall improvement over the baseline.

4.3 Qualitative Analysis

The qualitative behavior of the proposed variants is consistent with the quantitative results. Global FlowEdit tends to propagate the editing force to the whole latent representation, which can alter background texture and corrupt unrelated objects, license plates, and small text regions. The proposed variants constrain the correction to regions selected by attention or by velocity-gradient sensitivity. As a result, they better preserve background details and non-target objects while still modifying the target. The remainder of this section first inspects the spatial control signal itself — the hard and soft masks on a fixed sample, as the threshold is swept — and then illustrates the resulting edits on three representative editing scenarios drawn from the evaluation set, each picked to expose a different failure mode of the global baseline.

4.3.1 Spatial control signal: hard vs. soft mask across thresholds

Before discussing how each variant affects the final edit, it is useful to look at the spatial control signal itself — the per-pixel mask that gates the prompt-induced velocity difference. Figures 4.1 and 4.2 visualize, on the same `cat_dog_car` sample used in the next subsection, the binary hard mask for five values of the threshold τ and the continuous soft mask for five values of the sigmoid-bias parameter b . The same five values are swept in both cases: $\tau, b \in \{0.09, 0.12, 0.15, 0.18, 0.21\}$, matching the quantitative sweeps in Tables 4.1 and 4.2.

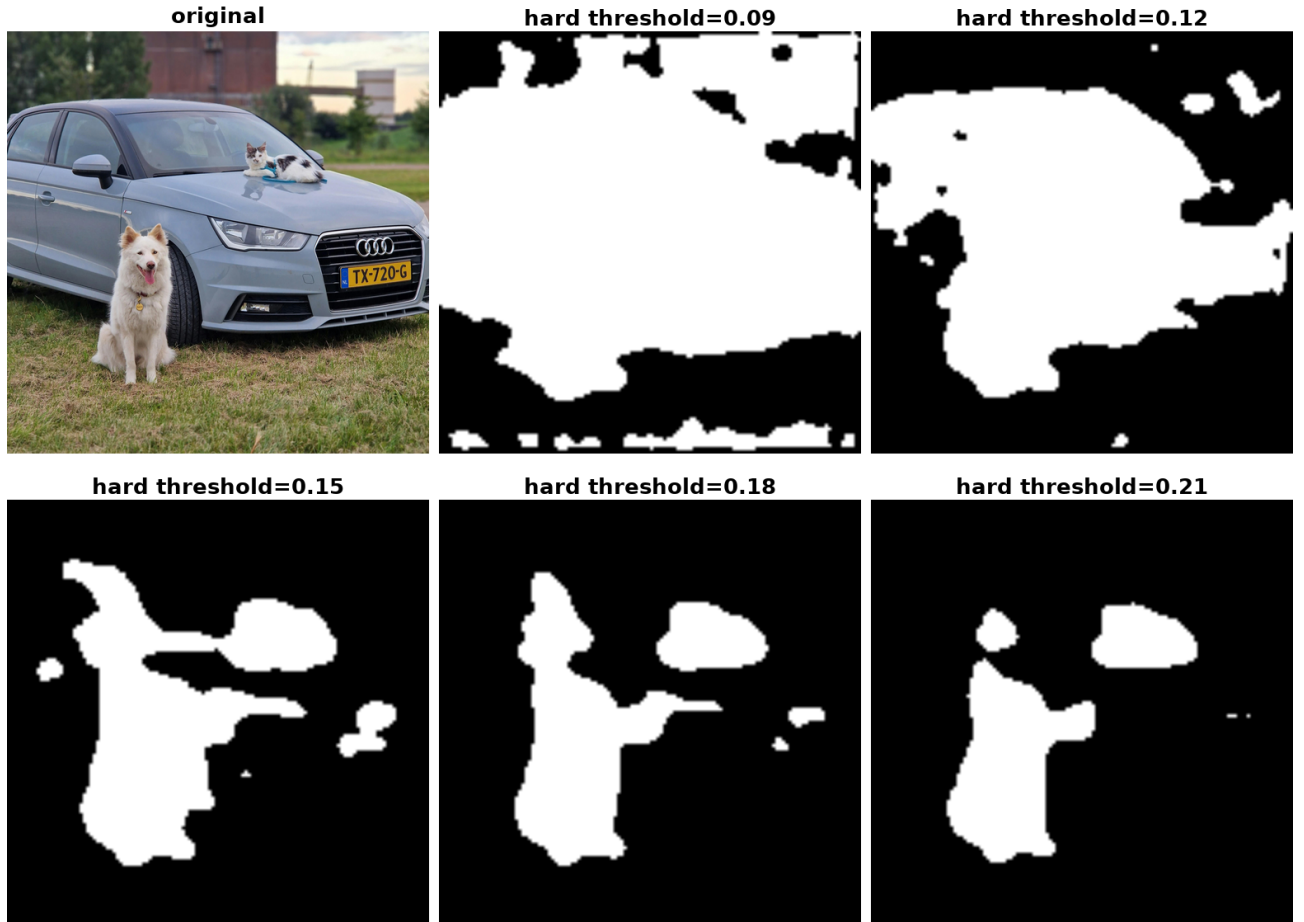


Figure 4.1. Hard-mask threshold sweep on the `cat_dog_car` sample. Top-left panel: source image; remaining panels: the binary hard mask at thresholds $\tau \in \{0.09, 0.12, 0.15, 0.18, 0.21\}$. White pixels are positions where the prompt-induced velocity difference is applied at full strength; black pixels are positions where it is fully suppressed.

The hard mask sharpens monotonically as τ grows. At $\tau = 0.09$ the mask covers a large fraction of the image, including parts of the grass, the sky, and the car body, so the resulting edit is close to the global FlowEdit baseline — the spatial selectivity is essentially absent. At $\tau = 0.12$ the mask still envelopes most of the car but pieces of the sky and grass already drop out. From $\tau = 0.15$ onwards the mask collapses onto the two semantically active regions of the target prompt: a dog-shaped blob on the left and a cat-shaped blob on the hood. At $\tau = 0.18$ those two blobs become more compact and the small spurious connections (above the dog’s head, around the licence plate) shrink. At $\tau = 0.21$ the mask is the tightest of the sweep — only a small core around each animal is retained — and parts of the dog’s silhouette and the cat’s torso are visibly cut off. The same monotonic trend underlies the quantitative sweep in Table 4.1: low thresholds approximate the baseline, mid thresholds achieve

the strongest preservation/alignment trade-off, and very high thresholds undershoot the edit because not enough velocity is delivered to the object regions.



Figure 4.2. Soft-mask sigmoid-bias sweep on the same `cat_dog_car` sample.

Top-left panel: source image; remaining panels: the continuous soft mask at sigmoid-bias values $b \in \{0.09, 0.12, 0.15, 0.18, 0.21\}$. Brighter pixels carry a larger fraction of the velocity difference; darker pixels carry less.

The soft mask shows the same trend in a continuous form. At small b the gate is broad and almost flat: the dog and cat regions are slightly brighter than the rest, but the entire scene receives a meaningful share of the velocity correction — the result, again, sits close to the baseline. As b grows the background fades and the dog and cat regions become progressively more dominant. At $b = 0.21$ the contrast between foreground and background is the largest of the sweep: the two animals are clearly highlighted while the grass, car body, and sky are visibly dimmed. Unlike the hard mask, the soft mask never reaches exact zero, so even at the strongest bias every pixel still receives a small fraction of the correction. This matches the quantitative behaviour in Table 4.2, where increasing b reduces LPIPS more aggressively than the

hard mask at the cost of a slight loss in CLIP and HPSv2.

Hard vs. soft on the same sample. Both masks select essentially the same spatial regions: across the entire sweep, the active region of the hard mask is the same dog-and-cat blob that the soft mask highlights. The structural difference is in the boundary. The hard mask is binary: a pixel is either fully edited or fully preserved, and the small attention noise translates into spurious holes inside the active region and small islands outside it (clearly visible at $\tau = 0.09$ and $\tau = 0.12$, where the white region is speckled with black holes). The soft mask is continuous and therefore has smoothly graded boundaries: the transition from full-edit to no-edit happens over several pixels, and the speckled noise that creates holes in the hard mask is reduced to a gentle local modulation of the weight. The practical consequences of this difference match what the quantitative results in Table 4.3 show:

- The hard mask, with its sharp on/off boundary, applies the *full* velocity correction inside the active region. This is what makes it effective for identity-changing edits such as cat $\rightarrow$ lion or dog $\rightarrow$ dalmatian: the active region is small but the correction inside it is strong enough to actually replace the object, which preserves the source–target alignment (CLIP) while the surrounding zeros protect the background (LPIPS).
- The soft mask never delivers the full correction at any pixel: even at the centre of the active region the weight is below one. As a consequence, identity-changing edits sometimes appear morphed (the rendered object is close to the target but not perfectly formed, as seen in the grocery-list “BACON”/“BRCON” example), but structural preservation is much stronger because every background pixel is moved only by a fraction of the velocity. The soft mask therefore wins on LPIPS at a small cost in CLIP and HPSv2.

These two behaviours — aggressive but tightly localized edits with the hard mask, attenuated but smoother edits with the soft mask — explain why the two methods are not interchangeable: each occupies a different point on the LPIPS–CLIP–HPSv2 trade-off curve, and the choice between them depends on whether the application prioritizes exact target rendering or maximal structural preservation.

4.3.2 Hard mask: simultaneous edit of two objects without background damage

Figure 4.3 shows the editing pair, which requires two simultaneous edits on the same image: the white dog standing next to the car must become a dalmatian, and the small cat on the hood must become a lion.



Figure 4.3. Hard mask vs. FlowEdit baseline. Left: source image. Middle: FlowEdit baseline. Right: hard mask ($\tau = 0.15$). Per-tile CLIP and LPIPS are printed above each panel. **Source:** “A blue-gray Audi car parked in a grassy area. A white dog sitting on the grass, next to the car. A cat laying on the hood of the car.” **Target:** “...A Dalmatian dog sitting on the grass, next to the car. A lion cub laying on the hood of the car.”

In the source image the car is a light-blue Audi, a white dog sitting on the grass to the left, and a small striped cat lying on the hood. The industrial background contains brick buildings and trees in soft focus.

The FlowEdit baseline (CLIP 36.51, LPIPS 0.2446) successfully replaces the white dog by a dalmatian, but *also* alters several regions that should remain unchanged: the cat on the hood disappears completely instead of being transformed into a lion, the licence plate is corrupted, the headlight geometry shifts. This is the canonical failure mode of global velocity blending: a strong target-prompt signal pushes the latent toward a state that satisfies the prompt only partially while sweeping away nearby objects and text that are not the focus of the edit.

The hard-masked variant (CLIP 39.11, LPIPS 0.1183) performs both required edits jointly: the dog becomes a dalmatian *and* the cat on the hood becomes a small lion, while every other region of the image is held essentially identical to the source

— the licence plate, the car body, the trees, and the grass texture are all preserved. The attention-derived mask correctly highlights both the dog region and the cat region (the two semantically active locations for the target prompt) and suppresses the velocity correction everywhere else, which prevents the background from drifting. The result is a strict Pareto improvement on this sample: lower LPIPS (−51.6%, much stronger preservation) *and* higher CLIP (+2.60 points, much stronger text alignment).

4.3.3 Soft mask: edit only the targeted word

Figure 4.4 shows the editing pair with the source image is a brown paper bag with a hand-written grocery list “−BREAD”, “−EGGS”, “−MILK”, and a small red London telephone-box figurine resting on top. The target prompt replaces only the first word: “BREAD” → “BACON”.

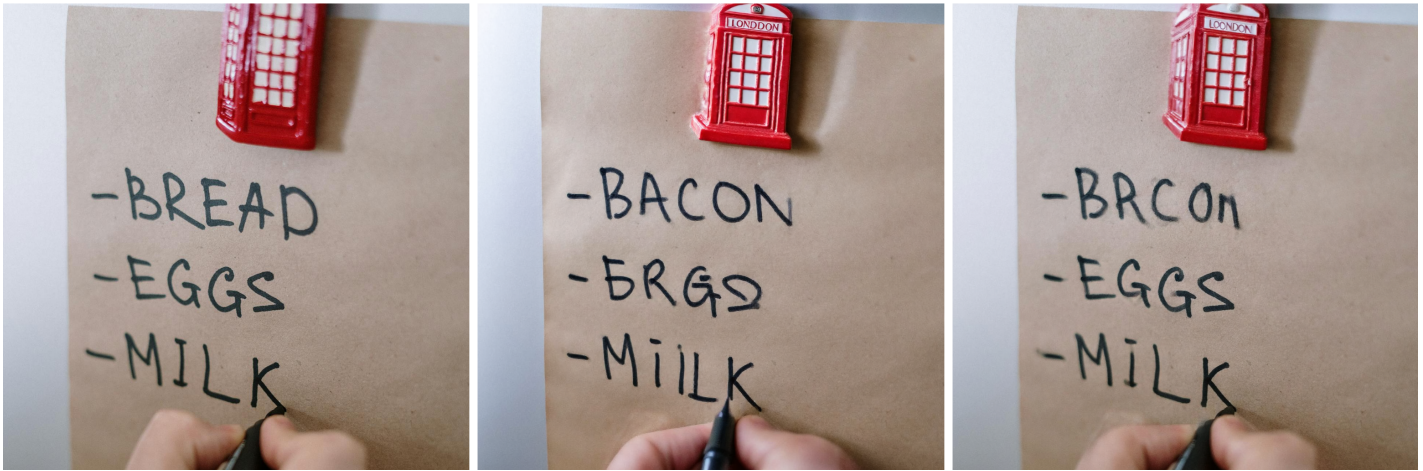


Figure 4.4. Soft mask vs. FlowEdit baseline. Left: source image. Middle: FlowEdit baseline. Right: soft mask ($b = 0.18$). Per-tile CLIP and LPIPS are printed above each panel. **Source:** “A grocery list writing on a piece of brown paper with a black marker, which includes BREAD, EGGS, and MILK. The grocery list is attached to a wall or refrigerator with a magnet. The magnet is red and white, possibly in the shape of a phone booth in London.” **Target:** “...which includes BACON, EGGS, and MILK...” (only the first list item differs from the source prompt).

The FlowEdit baseline (CLIP 43.21, LPIPS 0.1924) edits the targeted word correctly — “BREAD” becomes “BACON” — but it also *rewrites* the second word, which is now an illegible mix of letters (“bRGD”) that does not match the source “EGGS”. The third word is also slightly distorted (“MiLK” instead of “MILK”). The telephone-box figurine and the paper texture are preserved, but the unrelated hand-written lines have been damaged by the global velocity correction. This is the most

direct visual demonstration of the failure mode the proposed methods are meant to solve: even when the prompt mentions only one item, the global edit drifts into adjacent textual regions of the image.

The soft-masked variant (CLIP 43.87, LPIPS 0.0928) edits the first word to a letter sequence close to “BACON” (the rendered shape is somewhat morphed — visible as “BRCON” — because the soft mask attenuates rather than copies the velocity), but it keeps the second word as a clean “EGGS” and the third word as the original “MILK”. The telephone-box figurine and the paper bag texture are also preserved with markedly less local noise than in the baseline. The improvement is large on both axes: CLIP rises by 0.66 points and LPIPS drops by more than 50% (0.0928 vs. 0.1924). Even when the targeted word itself is not perfectly rendered, the overall edit is closer both to the prompt and to the source, because the corrections to non-targeted regions have been suppressed.

4.3.4 Velocity-Gradient Correction: cleaner text edit on a busy scene

Figure 4.5 shows the editing pair, which asks to replace the red sign “CAFE” on top of the gas-station awning by “CVPR”. This is a small but structurally constrained edit: a few large red letters in one rectangular region of the sky, with several cars, gas pumps, and a wide desert horizon that must remain identical.



Figure 4.5. Velocity-Gradient Correction vs. FlowEdit baseline. Left: source image. Middle: FlowEdit baseline. Right: Velocity-Gradient Correction ($\beta = 0.5$). Per-tile CLIP and LPIPS are printed above each panel. **Source:** “A gas station with a white and red sign that reads ‘CAFE’. There are several cars parked in front of the gas station, including a white car and a van.” **Target:** “...reads ‘CVPR’...” (only the sign text differs from the source prompt).

Both methods successfully render the new word “CVPR” on the sign. The differences are in everything that is not the sign.

The FlowEdit baseline (CLIP 39.81, LPIPS 0.1952) keeps the general layout but rearranges the cars under the awning, replacing the original sedans with a different mix of vehicles (a white van, a white SUV, and a different rear-most car), and erases the small secondary “HOT 5” banner that sits below the main sign in the source image. The mountain silhouette on the horizon is also softened. Each of these changes is small in isolation, but together they accumulate into a visible drift of the scene.

The Velocity-Gradient Correction variant (CLIP 40.20, LPIPS 0.1658) renders the same “CVPR” sign but keeps the cars in positions and tonalities closer to the source: the front car remains a light sedan in the same parking slot, and the second sedan stays in roughly the same position. The horizon and pumps are slightly cleaner, with fewer hallucinated details. The gain is more modest than in the hard-mask example — in numerical terms CLIP improves by 0.39 points and LPIPS by 0.029 — but it still represents a Pareto improvement on this sample: both metrics move in the favourable direction simultaneously. This is exactly the behaviour the quantitative sweep predicts for $\beta = 0.5$: a small, non-localized correction that buys back a fraction of the preservation loss caused by the global baseline.

Across all three figures the same pattern emerges: when the spatial signal is correctly localized, the corresponding variant achieves a strict Pareto gain over the baseline on the studied sample — lower LPIPS *and* higher CLIP at the same time — and the gain comes precisely from *not editing* regions of the image that the prompt does not require to change.

4.4 Chapter Conclusions

The experiments confirm that attention-based masking targets the main weakness of global velocity blending: the soft mask attains the strongest source-structure preservation, the hard mask gives the most balanced LPIPS / CLIP trade-off and dominates the baseline at low thresholds, while the attention-free Velocity-Gradient Correction yields only a marginal improvement — evidence that spatial localization, more than signal amplification, is what drives the observed gains.

CONCLUSION

This thesis addressed the problem of localized text-guided image editing with pre-trained flow-based generative models. The starting point of the work was the observation that global source-target velocity blending creates a trade-off between edit strength and source-image preservation. If the edit is too weak, the target object may remain unchanged; if it is too strong, background details and unrelated image regions are damaged.

The theoretical review showed that Rectified Flow and Flow Matching provide an efficient continuous-time formulation of generation, and that FlowEdit successfully adapts this formulation to inversion-free image editing. At the same time, the review identified a key limitation of existing methods: they lack an explicit spatial mechanism for deciding where the editing velocity should be applied.

To address this limitation, the thesis proposed Spatially-Adaptive Velocity Fusion. The method computes the difference between source-conditioned and target-conditioned attention maps, clips and normalizes this difference, and uses it as a spatial mask for the source-target velocity correction. Two variants were implemented: a binary hard mask and a continuous soft mask.

The empirical evaluation on 280 image-prompt pairs demonstrated that attention masking substantially improves structure preservation. The soft mask reduced LPIPS from 0.2129 for the FlowEdit baseline to 0.1415, while the hard mask at $\tau = 0.09$ reduced LPIPS to 0.1922 and *slightly improved* the CLIP Score relative to the baseline (35.11 vs. 35.09), demonstrating that attention masking can yield a Pareto improvement over the global baseline on (LPIPS, CLIP). The Velocity-Gradient Correction baseline produced only a minor LPIPS improvement and reduced HPSv2, showing that gradient amplification without spatial selectivity is not sufficient.

The main practical result of the work is a train-free modification of flow-based image editing that uses the internal attention structure of the model to localize edits. The main scientific result is empirical evidence that attention-derived masks can serve as an effective control signal for velocity-field modulation in text-guided image editing.

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